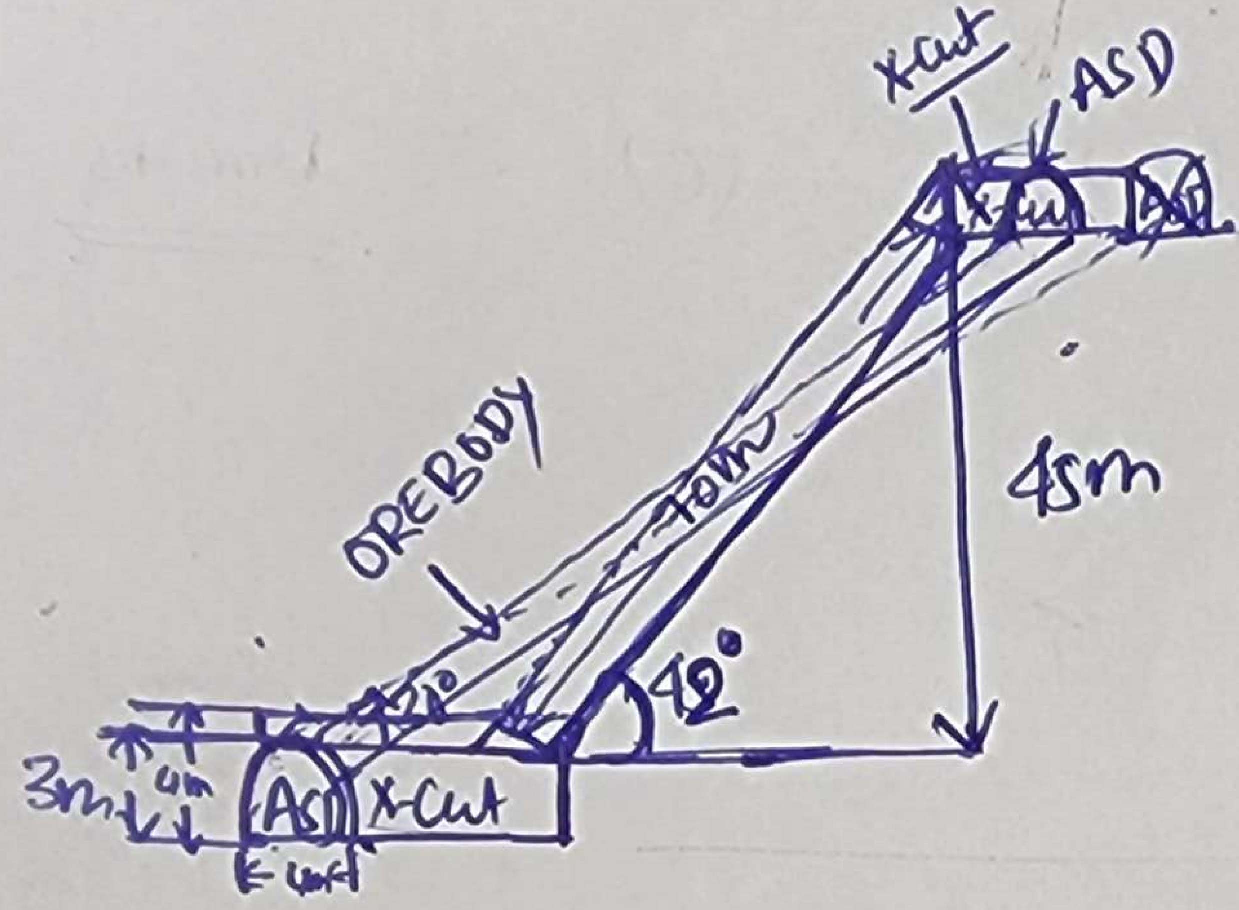
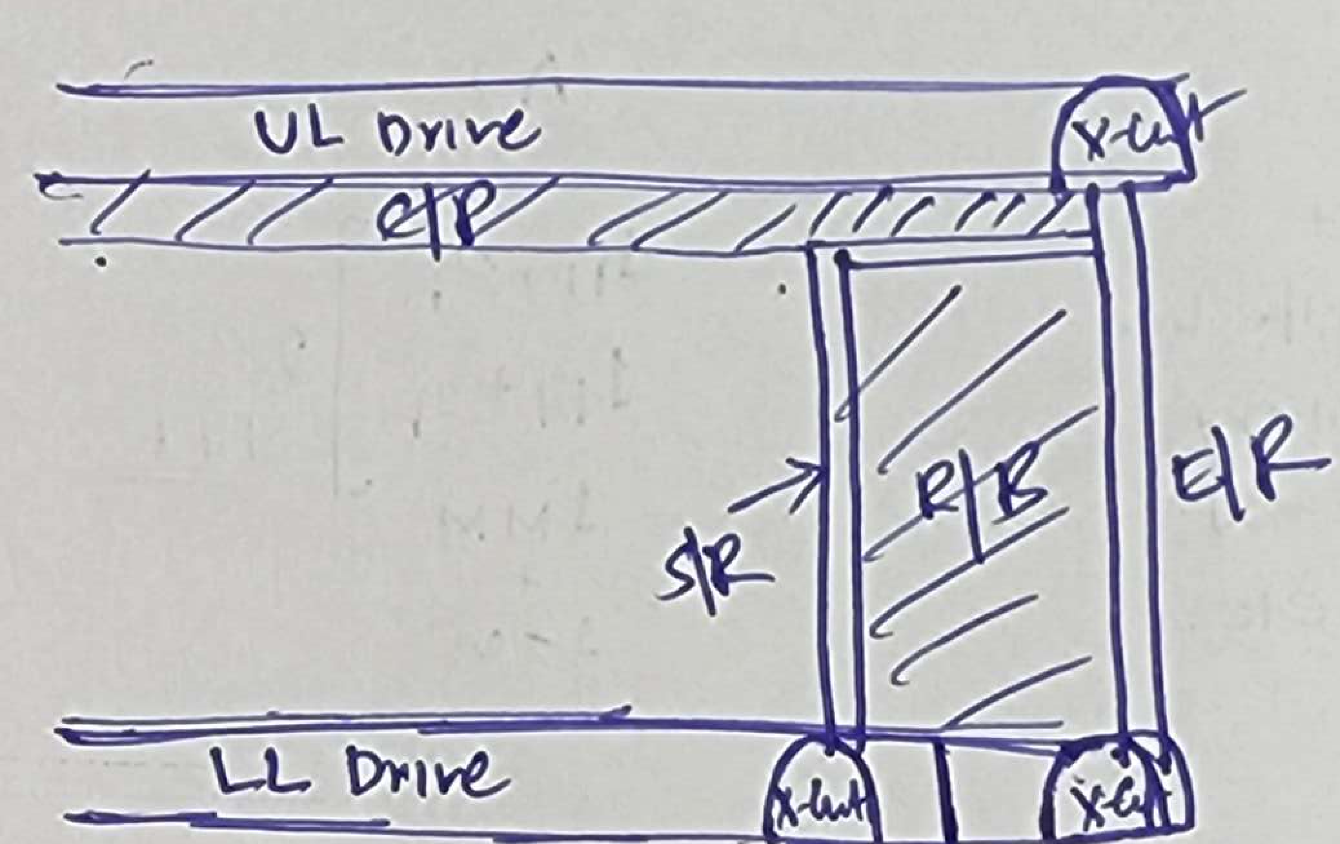
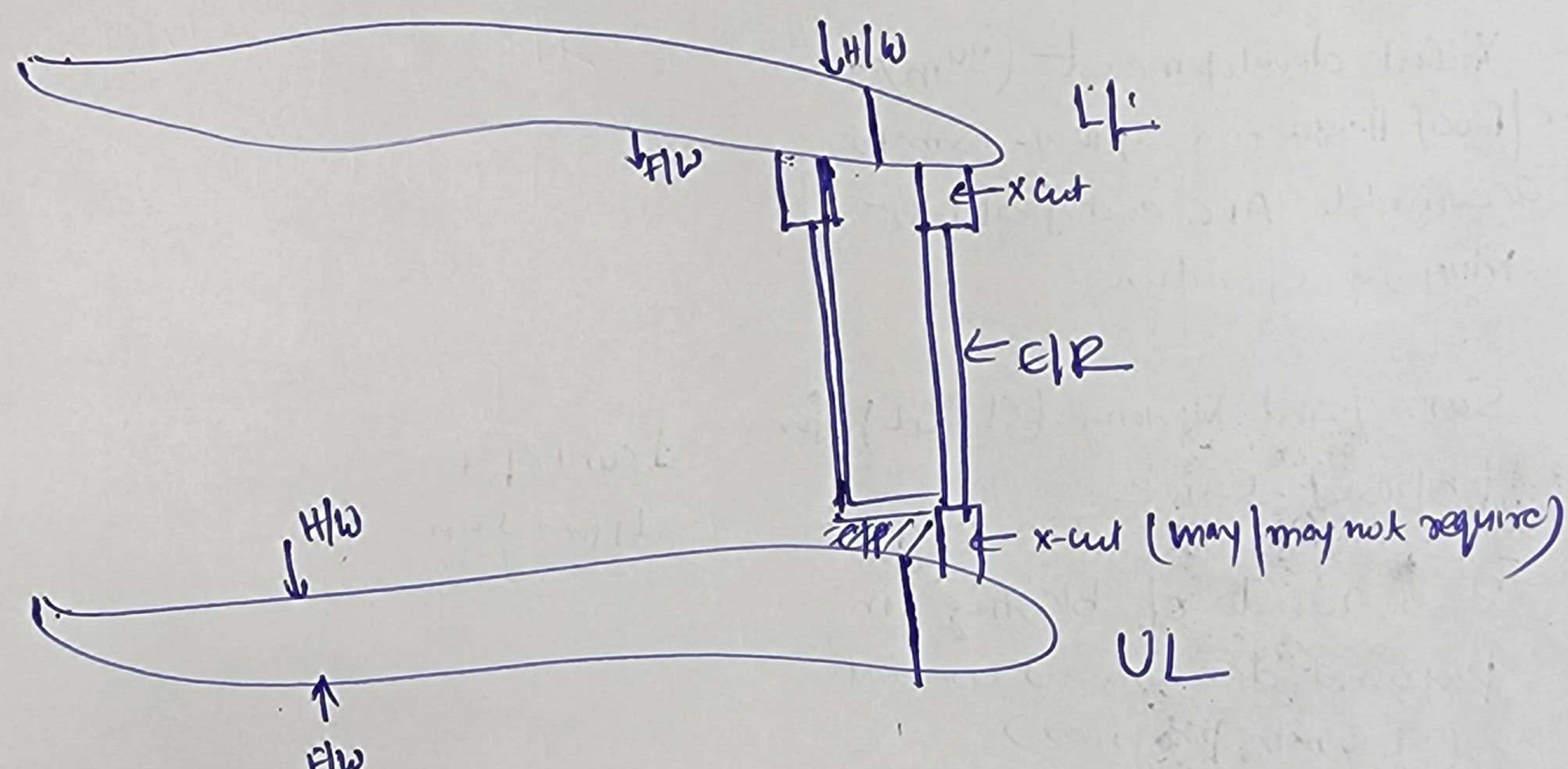
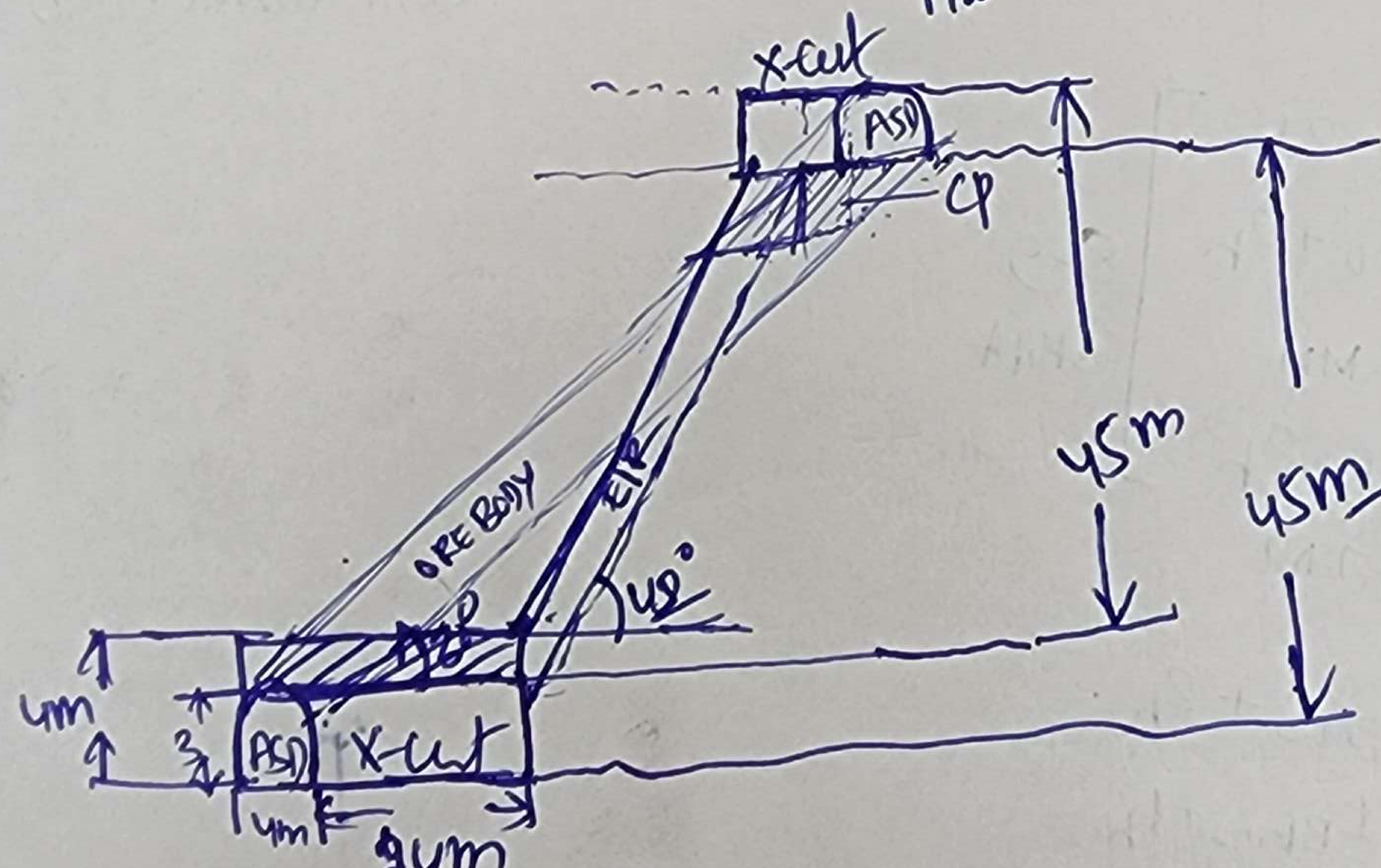
For 70m Raise development using ARC in given Condition :









Solution 4 (b) (i)

Steps in Installation:

(*) Angle guide fixing, Service guide fixing, Cage fixing, platform, Motor fixing, Hose reel Setup, Connection of Hose reel with Motor. etc., platform fixing, Canopy fixing.

Steps

(1) ~~These~~ preparatory work (extn of Water pipe line Compressed air pipeline, GEB, Duct pipe etc)

→ Planning & Survey

→ Survey + CL/GL Marking for X-cut (If required)

→ X-cut Development. (say 10m) + Roof Ht. to 4m to accomodate ARC and facilitate Mucking operation

→ ~~Installation of ARC~~ (*)

→ Survey + CL/GL for proposed Raise

→ 2-3 round of blasting in proposed direction ~~and~~ and Gradient

→ Installation of ARC (*)

→ ~~Drilling~~ face drilling + charging & blasting

→ Fume Clearance + WS & LD + face preparation + Repetation of cycle for 2-3 round of blasting

→ Service Guide rail extension of ~~face~~ to 2m length + face drilling (Incomplete)

(*) Service Guide is fixed with the help of Anchor bolt at 2 feet.

↳ face drilling is completed in next shift

↳ Repetation of cycle including ~~Guide~~. Service guide Rail extn.

↳ Mucking after 3-4 round of blasting

↳ Development Survey for CL and GL extn. along with measurement of opening size after 15-20m face advancement

↳ Uninstallation of ARC

↳ fitting of Water Ladder in the floor to make it operational.

Manpower distribution plan :

Activity	Shift (A)	(B)	(C)	Remarks
1. Preparatory work Water pipe line + Compressed air pipe line Vent. duct + Auxiliary Fan Installation + GEB + lighting etc.	1PF + 2h 1EI + 2h 1MM + 1TM 2-3 Shift			
2. Survey and Marking (CL&GL) for X-cut	1sur + 2h + 1TM + 1MM			
3. X-Cut development (24m) + back/Roof Heightening upto 4-4.5m to accomodate ARC and facilitate Mucking operation				Follow manpower distribution plan for Drift/Drive
4. Survey and Marking (CL&GL) for proposed Raise	1sur + 2h + 1TM + 1MM			
5. 2-3 rounds of blasting (6') in proposed direction and Gradient (4-5 mtr. progress)				Follow manpower distribution plan as discussed in Conventional Raising
6. Shifting & Installation of ARC	1Mt + 2h 1JHO + 2h 1MM 1PF + 2h 8-9 Shift 1TM			
7. Face drilling + Charging & Blasting + FC ~~FC + WS&LD + FP~~.	1JHO + 2h. 1Blaster + 1h 1MM + 1TM			
8. ~~FC +~~ WS & LD + FP + Face drilling (Incomplete)	1TM + 1MM + 1JHO + 2h			Repetation of step 7, 8, 9 for 2-3 rounds of blasting.
9. Face drilling (Complete) + Charging & Blasting	1JHO + 2h + 1Blaster + 1h + 1MM + 1TM			
Repetation 10 Service Guide Rail extn. + Face drilling (Incomplete)	1JHO + 2h 1TM + 1MM			

(4'&6')

	A	B	C	Remarks
11.	Face drilling complete + Charging & Blasting + FC	1 JHO + 2h 1 Blaster + 1h 1 TM + 1 MM		
12.	WS & LD + FP + Face drilling (Incomplete)	1 TM + 1 MM + 1 JHO + 2h		
~~13~~				
13.	Face drilling (Complete) + Charging & Blasting + FC	1 JHO + 2h 1 Blaster + 1h 1 TM + 1 MM		
14	~~Same~~. WS & LD + FP + Service Guide extn + Face drilling (Incomplete)	1 TM + ~~1 MM +~~. 1 MM + 1 JHO + 2h		Repeatation of step 10, 11, 12 & 13
15	Mucking after 3-4 round of blasting	—		
16	Development Survey (CL & GL extn + Opening Size Measurement) after 15-20 m Face progress / development	1 Sur + 2h		
17.	Uninstallation of ARE	1 Mech + 2h + 1 TM + 1 MM	15 shift	
18.	fitting of Wooden Ladder	1 TM + 2h + 1 MM + ~~1~~	15 Shift	

⇒ Total days required to Complete it

Activity	Shift
1	2-3 shift
2	1 Shift
3	12 shift
4.	1 shift
5.	3+3 shift
6.	8-9 shift

Activity	Shift
7, 8, 9, 10 11, 12, 13	4 blast in 7 shift] per day 1 blast of $\frac{4'+6'}{2} = 5'$ (Considering 2 shift working in Raising below face) (50 days for 65 m development)
15.	HR
16.	1 shift
17.	15 shift
18.	15 shift

Total days required = ~~90~~ 81 days for 70 m development

~~So,~~ So, days required for 1 m development = $\frac{\cancel{90}\,81}{70} \simeq \cancel{1.28}$ days/m

1.1 days/m

Stope Block Development and its Completetion

Q5) (a) Types of the excavation :

1) Initial X cut which will act as drift later on.
2) Across X cut
3) Development of end raise / ventilation raise
4) Development of drive (ASD, Advanced Strike drive) (100 m)
5) Drift development (100 m)
6) ASD development (next 100 m)
7) next end raise / ventilation raise
8) drift development (next 100 m)
9) E/R – Drift – X cut
10) 2 Stope raises have to be developed (5 m below level)
11) ramp up development (after 2–3 rounds of slicing)
12) development of ore transfer raise.

9

(ii) Days required :

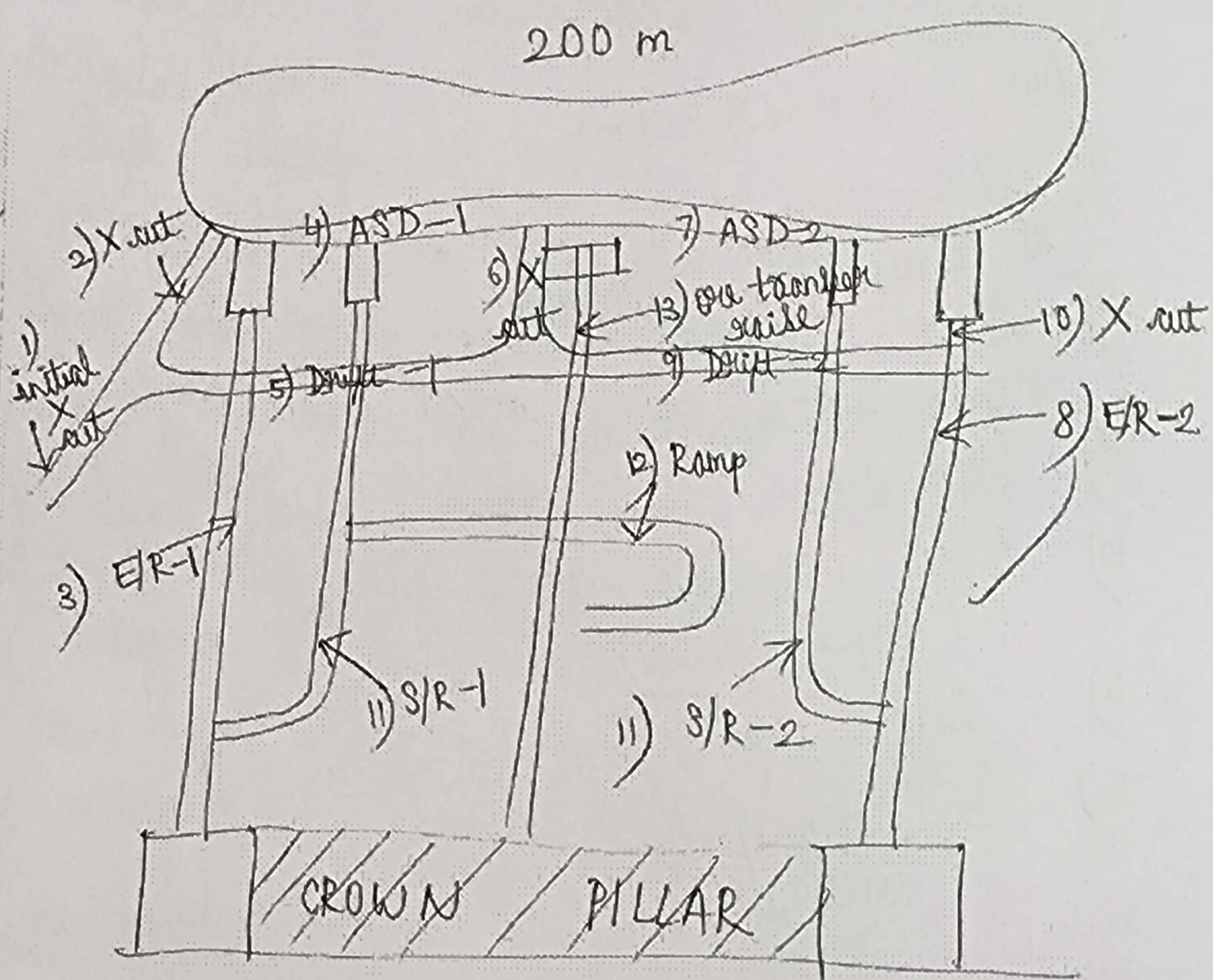
Type of excavation in sequence	Days required
1) initial X cut (40 m, 4.5 m X 3.2 m)	20 days $\left(\frac{40}{2.9} + \frac{10}{100} \times \frac{40}{2.9}\right)$

2) End raise – 1 (70 m, 45°, 2m X 2m)	100 days
3) ASD – 1 (100 m, 4·5m X 3·2m)	40 days
4) Drift – 1 (100 m, 4·5m X 3·2m)	40 days
5) access X cut (35m, 4·5m X 3·2m)	20 days
6) ASD – 2 (100 m, 4·5m X 3·2m)	40 days
7) E/R – 2 (70 m, 45°, 2m X 2m)	100 days
8) Drift – 2 (100 m, 4·5m X 3·2m)	40 days
9) X cut	20 days
10) S/R – 1 & S/R – 2 (stope raises)	90 days
11) Ramp up	20 days
12) Ore transfer raise	100 days

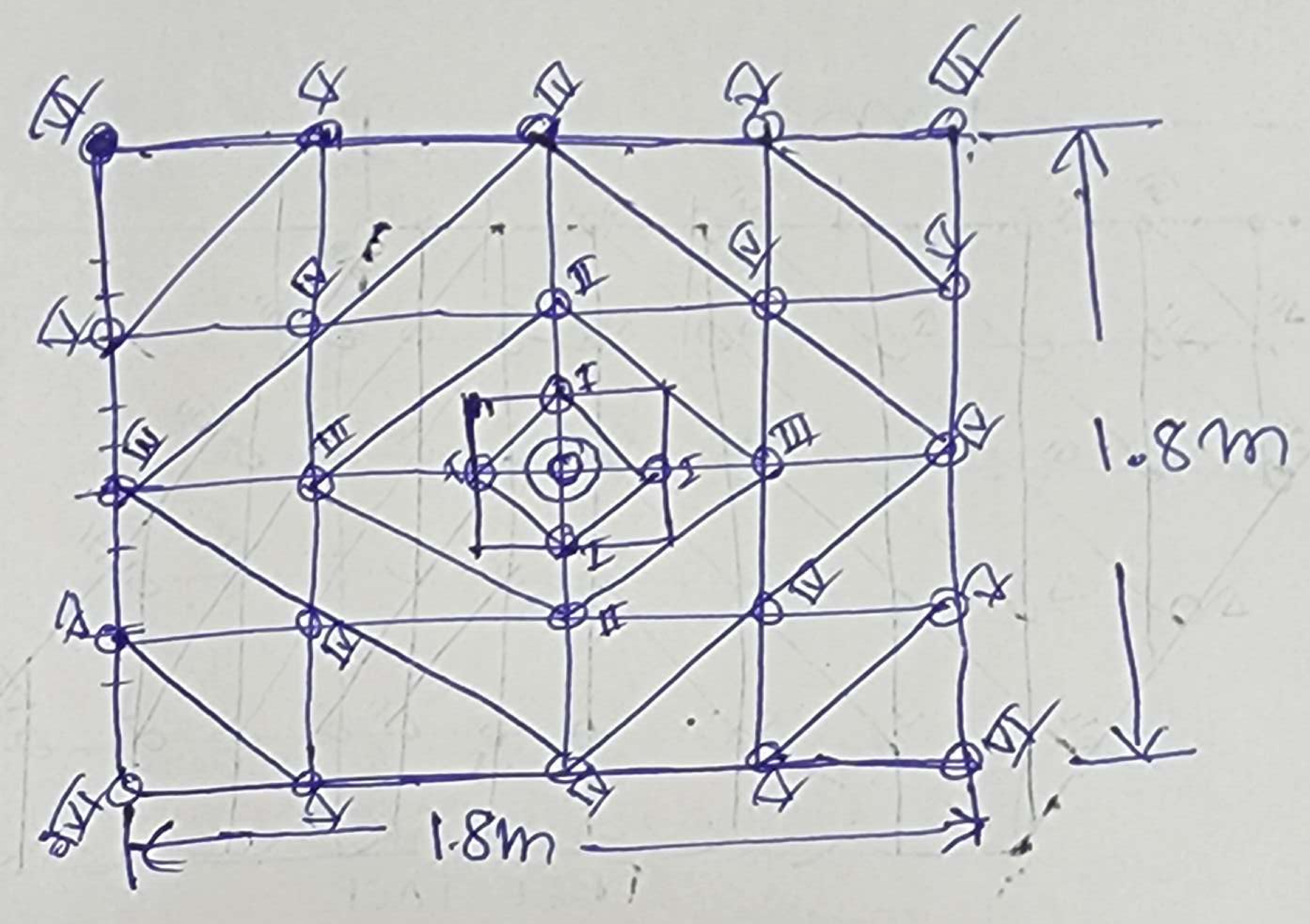
Total time = 2 ~~months~~ years

∵ we need to consider double the time because there may be chances of unforeseen circumstances and breakdown of machineries,

hence total time required for completing block development will be 4 years.



Drilling & Blasting pattern of Jack hammer Face (2m x 2m) : Raising (Conventional & ARC)



Note: Roman letter shows sequence of blasting

Total No. of Holes = 29 + 1R

Total No. of Blast holes = 28

Total Meterage (4 feet depth of per holes) = ~~28 X 1.2~~ = 29 X 1.2 + 1 X 1.2 .

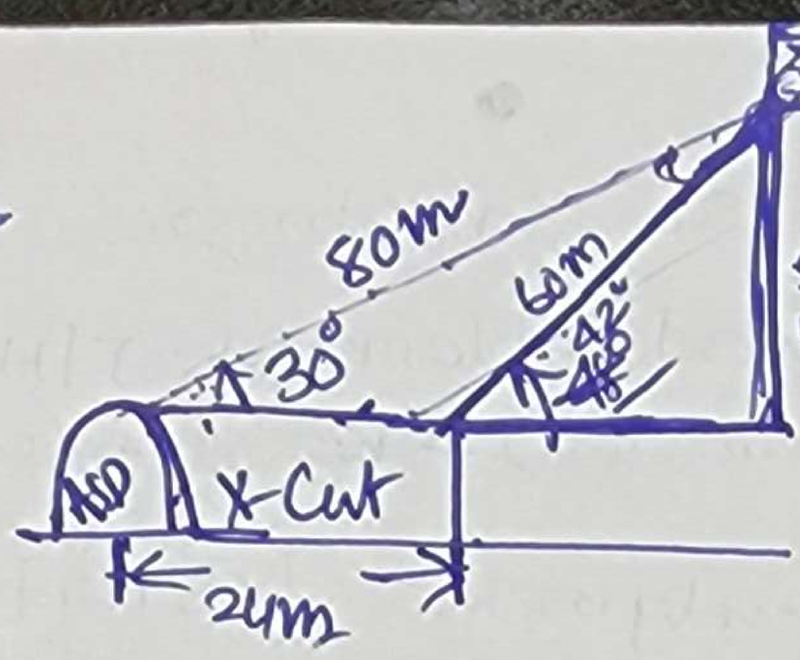
=

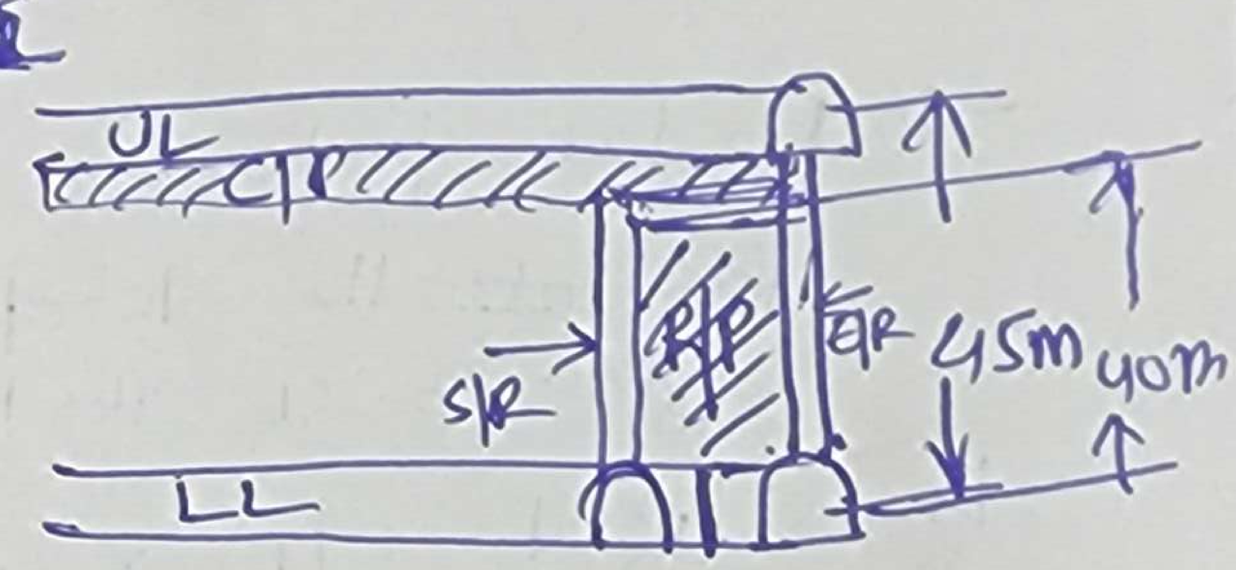
Total Explosive per round = 28 X .7 kg = 20 kg

Total Tonnage per round of Blasting = 2 X 2 X 1.2 X .9 X 3

= 13 tonnes.

CONVENTIONAL RAISING





Ⓐ Steps Involved:

1. Preparatory Work : Compressed Air pipe line, Water pipe line, Auxiliary fan + vent. duct, lighting. etc

2. Survey and Marking of S|R X-cut (if required)
3. Development of S|R X cut in proposed direction as marked by Surveyor.
4. Survey and Marking of S|R for direction and Gradient.
5. Face Drilling using J|H drill m|c with 2'|4' & 6' drill rod in (Burn cut pattern)
6. Charging & Blasting
7. Fume clearance
8. Water spraying + Loose dressing + Face preparation
9. Repetition of steps 5, 6, 7, 8 for next three round of blasting
10. partial Mucking using LHD|LPDT &
11. Survey + Marking + fixing pegs
12. platform preparation for next round of drilling atleast 2m behind the face. + Rope ladder fixing in the floor
13. Face Drilling + side support + Charging holes for platform + ladder + Blasting (after dismantling the platform) + fume clearance + WS + LD + Face preparation + platform preparation
14. After 2-3 rounds of blasting partial Mucking is done.
15. Development Survey after 10-15 m progress : checking C|L & G|L and extending C|L and G|L in the proposed direction + & measuring the face size (Height and width)
16. extending the Rope ladder as face advances (Generally it is kept just behind the platform location.)

→ Manhole preparation after 20-25 m progress of the face ~~to~~ for ~~facilitate the~~ Keeping Wooden planks + J/H M/C (Face Marking + Drilling + C&B (after platform dismantling) + ~~ws~~ FC + WS + LD + Face preparation

→ S/R to E/R X-cut development (5 meters) in Horizontal plane

Survey and Marking C/L & G/L in proposed direction and Gradient

Next 2-3 round of blasting with the help of platform

~~#~~ # Rest without platform

Muck Scraping is an additional step in the X-cut development ~~to which~~ which takes more time for face preparation for next round of drilling & Blasting)

(B) Manpower distribution plan Activity wise : SN

(C) Days required to Complete the Raise : SN 92 days

(D) Eqpt / Machinery deployed : SN

(E) Drilling and Blasting pattern : From ARC Note

(F) Drilling Mtrge for round of blasting : From ARC Note

(G) Explosive Consumption per round of Blasting : from ARC Note

(H) Tonnage produced per round of blasting : from ARC Note

(I) Powder factor Obtained : from ARC note

(J) Ventilation ckt for ventilating the face : from ARC Note

Development of Horizontal excavation (Such as : Drift | Drive | X-Cut | Raise/Decline etc) ~~using~~ with Drilling and Blasting excavation Method :

↳ ~~planning~~ Survey of the proposed Location(s) as per the plan and ~~[illegible]~~ Marking Central line and Gradeline (for Direction and Gradient)

↳ **Drilling** with drill jumbos or Jack hammer in the proposed direction and gradient (Drilling pattern : Burn cut for hard rock, Dia of hole .... Length of hole 3.2 m (drill jumbo) ; 8 feet | 2 m (Jack hammer), face size 4.5m X 3.2m or 3X3 m

↳ **Charging & Blasting** : Charging with emulsion explosive in the form of Cartridge, Electric delay detonators are used, Blasting with the help of exploder from a safe distance atleast one right angle from the face direction. Blasting is done at the end of the shift. during blasting Auxiliary fan should remain OFF.

↳ ~~fume clearance :~~

↳ **fume clearance** : fume clearance is done with the help of Auxiliary forcing fan | Combination of forcing & exhaust fan.

↳ **Water Spraying & Loose Dressing** : Water spraying & Loose Dressing is done Manually. Water spraying suppresses the dust and loose dressing operation helps in separating loose rock generated after blasting in the back and sides of the face.

↳ **Loading & Hauling of Blasted material** : Loading ~~operation~~ & Hauling operation ~~is done~~ is carried out using LHD & LPDT combination in Trackless Mining System whereas Tub | Granby Car & Rocker Shovel | Cavo loader (pneumatically operated) used in Track bound Mining System. Loaded Material is dumped near by u/g Void ~~or~~ (if available) otherwise hoisted to surface and dumped in ~~[illegible]~~ waste dump yard.

↳ face preparation | ~~Bod~~ Ready for next round of drilling : finding sockets and Misfire (if any) and cleaning the holes.

↳ Support of back & Side (if required) : Support of back as per the Support plan (SSR) framed by the Mine with the approval of DGMS. Generally, Roof bolt of 2m is used to Support the back. in the grid of 1.8m X 1.8m. (Cement + Sand + Water) is used as a grouting material. full Column grouting is done.

↳ Repeatation of the process (Drilling → Charging & Blasting → fume clearance → Water spraying & Loose dressing → Loading & Hauling of blasted rock → face Ready → Support)

↳ Development Survey:

fixing Centre peg & Grade peg and extending the Centre Line and Gradeline :— In order to Maintain direction and Gradient of the excavation | Development type, after ~~15-20m~~ face advancement around 20-25m (In the Case of Drill Jumbo face and around 10-15m (in the Case of Jack hammer face), Centre pegs and Grade pegs are fixed by the Surveyors in the back and Sides of face respectively. a set of New Centre pegs (2 Nos.) and Grade pegs (2+2=4 Nos) Set in ~~Shifted~~ established, after 20-25m face advancement.

~~Development Survey~~ :

Measuring Size of the opening (HXW) : In order to assess the Size of the opening as per the actual Size, Development Survey is done. it helps in monitoring the overbreakage or underbreakage. (both are undesired)

↳ Drainage : Drainage is to be provided to drain the water from the face. (for Decline and Ramp down, face pump is required during drilling and charging), for drift | drive, it is developed in up-gradient (1 in 200) and drain is required. for Ramp up and Ramp down, No need of ~~drainage~~ | Drain, for Decline: Drainage is made at the one Side of the floor.

↳ Extension of Auxiliary Services : Water pipe line, Compressed air pipe line, Electrical Cable, Gate end box, Manhole Ventilation duct. are extended at regular Interval as face advances.

↳ Extension of Auxiliary services : Water pipe line, Compressed air pipe line, Electrical Cable, Gate end box, Manhole, Ventilation duct, peg set up, Loading bay are extended at regular Interval as face advances.

Other activities	No. of Shift required
→ Survey and Marking	1 Shift
→ preparatory Work : Compressed Air pipe line, Water pipe line Installation of Auxiliary Fan + duct + Electrical Cable Laying, GEB.	3 Shifts
→ Loading bay	11 shift
→ Manhole + support holes	1 Shift @ 20m, for 100m . 5 Shift
→ peg set up + ~~size~~ Opening size measurement	1 shift @ 20m, for 100m. 5 Shift

Main Activities

Drilling + charging + blasting + WSLD + fume clearance + Loading & Hauling	3 shift per round of blasting
Support (back + side)	. 2 Shift @ every 3 round of blasting

→ Total Shift needed for (100m development) $= \frac{\cancel{100}}{\cancel{288}} \cdot \frac{100}{3.2 \times 0.9} \times 3 + \frac{100}{3.2 \times 0.9} \times \frac{2}{3}$

$+ 1 + 3 + 11 + 5 + 5$

$= 104 + 23 + 25 = 152$

→ Total Shift after ~~taking~~ Considering B/d time and unforeseen situations $= 152 + 152 \times 5\%$

$= 152 + 7.6 = 159.6 \simeq \underline{160}$

~~No. of days~~ → In days, $\frac{160}{3} = 53$ days ~~$= \frac{160}{3}$~~ =

→ In month ; $\frac{53}{25} = 2.12$ months

Table 14.6 Comparison of Advantages and Disadvantages of Underground Methods

	Unsupported				Supported			Caving		
Characteristic	Room-and-Pillar	Stope-and-Pillar	Shrinkage	Sublevel	Cut and Fill	Stull	Square-Set	Longwall	Sublevel Caving	Block Caving
1. Mining cost	20%	10%	45%	20%	55%	70%	100%	15%	15%	10%
2. Production rate	Large	Large	Moderate	Large	Moderate	Small	Small	Large	Large	Large
3. Productivity	High	High	Low	High	Moderate	Low	Low	High	Moderate	High
4. Capital investment	High	Moderate	Low	Moderate	Moderate	Low	Low	High	Moderate	High
5. Development rate	Rapid	Rapid	Rapid	Moderate	Moderate	Rapid	Slow	Moderate	Moderate	Slow
6. Depth capacity	Limited	Limited	Limited	Moderate	Moderate	Limited	Unlimited	Moderate	Moderate	Moderate
7. Selectivity	Low	High	Moderate	Low	High	High	High	Low	Low	Low
8. Recovery	Moderate	Moderate	High	Moderate	High	High	Highest	High	High	High
9. Dilution	Moderate	Low	Low	Moderate	Low	Low	Lowest	Low	Moderate	High
10. Flexibility	Moderate	High	Moderate	Low	Moderate	High	High	Low	Moderate	Low
11. Stability of openings	Moderate	High	High	High	High	Moderate	High	High	Moderate	Moderate
12. Subsidence	Moderate	Low	Low	Low	Low	Moderate	Low	High	High	High
13. Health and safety	Good	Good	Good	Good	Moderate	Moderate	Poor	Good	Good	Good
14. Other	Highly mechanized, caves with pillar recovery, good ventilation	Mechanized, fair ventilation	Gravity flow in stope, labor-intensive, simple method	Gravity flow in stope, mechanized, large blasts, good ventilation	Gravity flow in stope, mechanized, requires backfill	Gravity flow in stope labor-intensive, simple method	Gravity flow in stope, labor-intensive, high timber cost	Highly mechanized, continuous, rigid, expensive moves	Mechanized, draw control critical	Gravity flow in undercut, low breakage cost, good ventilation, draw control critical

Table 14.7 Selection Chart for All Underground Mining Methods

Deposit Shape	Deposit Orientation	Deposit Thickness	Ore Strength	Rock Strength	Applicable Method(s)
Tabular	Horizontal, flat	Thin	Strong	Strong	Room-and-pillar mining, stope-and-pillar mining
			Weak, strong	Weak	Longwall mining
		Thick	Strong	Strong	Stope-and-pillar mining
			Weak, strong	Weak	Sublevel caving
	Vertical, steep	Thin	Strong	Strong	Shrinkage stoping, sublevel stoping
			Strong	Weak	Cut-and-fill stoping, square-set stoping, stull stoping
			Weak	Strong	Square-set stoping
			Weak	Weak	Square-set stoping
		Thick	Strong	Strong	Shrinkage stoping, sublevel stoping
			Strong	Weak	Cut-and-fill stoping, sublevel caving, square-set stoping
			Weak	Strong	Sublevel caving, block caving, square-set stoping
			Weak	Weak	Sublevel caving, block caving, square-set stoping
Massive	—	—	Strong	Strong	Shrinkage stoping, sublevel stoping
			Weak	Weak, strong	Sublevel caving, block caving, square-set stoping

Sources: Modified after Peele, 1941; Lucas and Haycocks, 1973; Thomas, 1978.

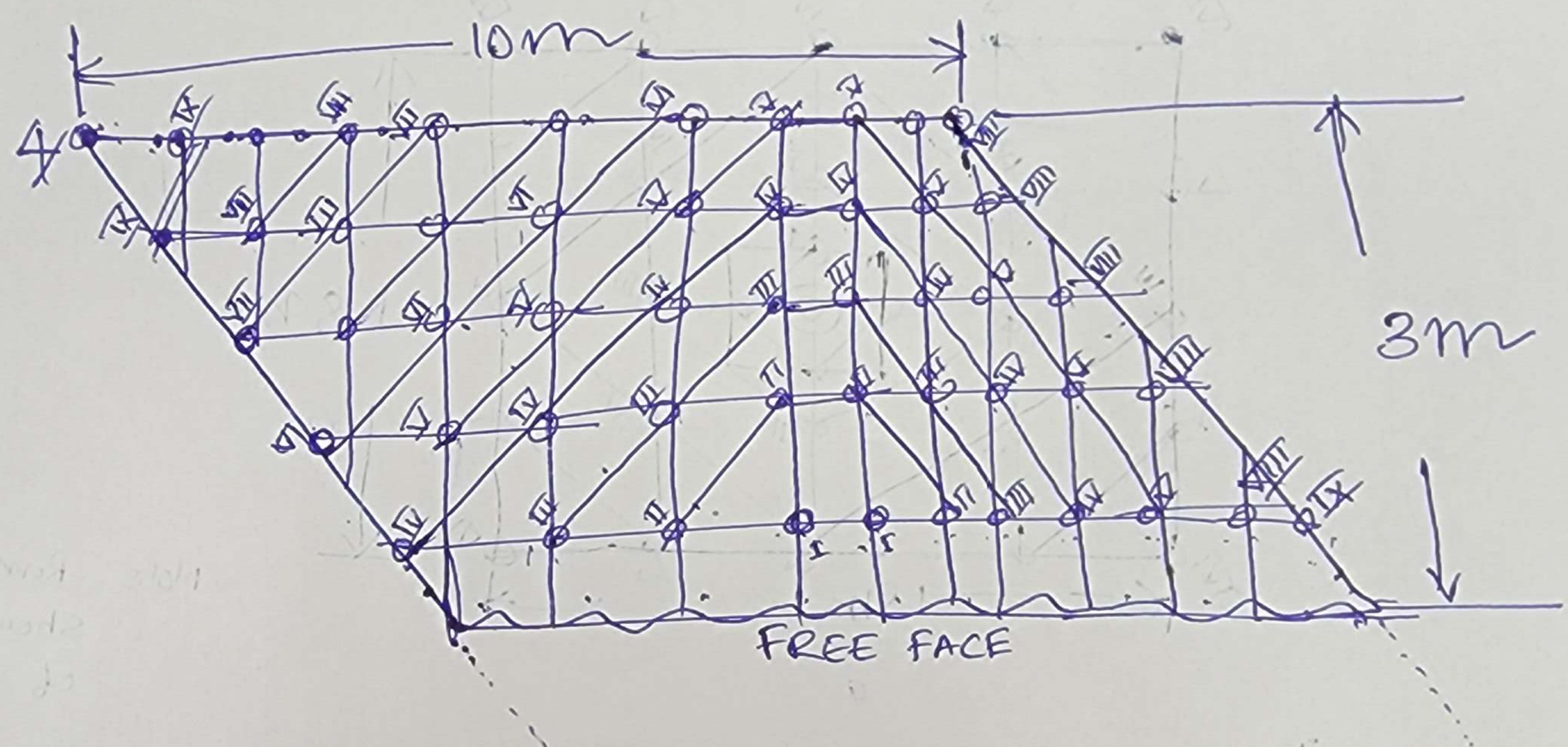
14.2.3 Novel Versus Traditional Methods

Because their characteristics and applications differ significantly, it is more difficult to compare traditional methods with novel methods than surface with underground methods. By their very nature, novel methods have been devised to meet the unique requirements of a particular mineral deposit or to employ the unusual features of a new technology. In addition, they may have been utilized less than traditional methods and may not have a suitable track record for cost estimation. Accordingly, they will ordinarily be more risky than traditional mining methods.

The characteristic of the mineral deposit that is most discriminating in the selection of a novel mining method is the mineral commodity itself (see Table 13.1). Accordingly, we summarize the most likely possibilities for each of a series of deposit types:

Slice drilling and Blasting pattern (HCF)

~~Face size~~



Total No. of Blast holes ≃ 50

Total Mtge per round of blasting ≃ 50X3.2 ≃

Explosive per hole ≃ 2.7 to 3 Kg

Total Explosive per round of blasting ≃ 50 X (2.7 - 3) ≃
≃ 150 kg

Total Tonnage per round of Blasting = 10X3X3.2X3
≃ 270 tonnes

Note: Roman Letter shows sequence of Blasting.

S.No 9 (*) Level interval is 45m, Dip of the orebody is 30°, thickness (plan) 9m. etc

A block of HCF of length 200m in between two levels ~~particular~~ level is to be developed ~~using~~ with a ~~combination of~~ Trackless ~~and track bound~~ mining system. (*) Describe the following:

1) Types of excavation required for to develop
2) Steps involved in its development
3) Time required to complete the block development
4) plan | sections | Drawing, wherever needed.

5)

Sl. No 11

A developed block of 200m for HCF is to be stoped with trackless mining system; ~~W~~- plan width of the orebody is 9m. Dip = 30°; Level Interval 45m.

Describe the followings

1) steps involved in the stoping process
2) Time required for its completion
3) Manpower distribution plan shift wise
4) plan | sections | figure wherever needed
5) Drilling MTs per round of blasting
4) Explosive consumption per round of blasting
5) Drilling pattern
6) Blasting pattern

assume your condition, wherever needed.

No.

1 Steps

~~Details of~~

- ↳ planning and Survey (~~stope plan~~ preparation)
- ↳ ~~updated plan with~~ Generation of preparation stope plan
- ↳ Heightening of ~~roof~~ drive/back (3.2 m to 5.0m) across the strike
- ↳ widening of gallery / drive from 4.2m to 9.0m Across the strike
- ↳ Ht + widening upto the length of the block (~~both~~ one side upto S/R ~~...~~) along the strike
- ↳ Ht. of stope Raise
- ↳ Stope Survey and marking of stowing line upto 3m
- ↳ back filling (Waste rock + Mill tailing)
- ↳ Ht + widening upto the length of the block (other side upto S/R) along the strike
- ↳ Ht. of stope Raise
- ↳ Stope survey and marking of stowing line
- ↳ Back filling (Waste Rock + Mill tailings)
- ↳ Roof Ht. of Access cut and making ramp to access the stope ~~Unit operation~~
- ↳ Stoping (slicing) of 3m ht with 9m width ~~along the strike~~ of sub block-1 along the strike up to S/R
- ↳ Back fill of sub-block-1
- ↳ stope survey and marking of stowing line

Date:

No.

Avinash (RPS) 24/4/2023

Integration of 3D-Numerical Modelling and InSAR Techniques for developing a ~~pre~~ prediction model land surface deformation prediction Model.

Mining induced land surface deformation prediction Model

↳ Design of literature Review

↳ Slicing of sub-block-2 upto the ht of 3m and width 9m along strike upto S/R

↳ ~~Be~~ Back filling of sub-block-2

↳ Stope Survey and marking of stowing line

↳ Repetation of the process upto the crown pillar

↳ Access of stope from ramp-up ~~once~~ for next round of slicing (After every three round of slicing Access would be from next Access X cut from the ramp up

No.

Unit operation in per round of Blasting,

↳ Slice Drilling → Charging & Blasting → fume clearance → Water spraying loose → Loading & Hauling → ~~Drilling~~ support drilling (back and walls) + Roof bolting (back + walls)

Times

Sl.No 9 ⊛ Level interval is 45 m, Dip of the Orebody is 30°, thickness (plan) 9 m. etc

A block of HCF of length 200m in between two levels ~~particular~~ level is to be developed ~~using~~ with a ~~combination of~~ Trackless ~~and track bound~~ mining System. ⊛ Describe the following:

1) Types of excavation required for its development
2) Steps involved in its development
3) Time required to complete the block development
4) plan | Sections | Drawing, wherever needed.
5)

Sl.No 11

A developed block of 200 m for HCF is to be stoped with trackless mining system; ~~in~~ plan width of the orebody is 9 m. Dip = 30°; Level Interval 45 m.

Describe the followings:

1) steps involved in the stoping process
2) Time required for its Completion
3) Manpower distribution plan shift wise
4) plan | Sections | figure wherever needed.
5) Drilling Mts per round of blasting
4) Explosive Consumption per round of blasting
5) Drilling pattern
6) Blasting pattern

assume your condition, wherever needed.

No.

1 Steps

~~the etc of~~

↳ planning and Survey (~~stope plan~~ preparation)

↳ ~~updated plan with~~ Generation of stope plan preparation

↳ Heightening of ~~roof~~ drive/back (3.2 m to 5.0 m)

↳ widening of gallery / drive from 4.2 m to 9.0 m Across the strike across ~~along~~ the strike

↳ Ht + widening upto the length of the block (~~both~~ one side ~~to~~ upto S/R) along the strike

↳ Ht. of stope Raise

↳ Stope Survey and marking of stowing line upto 3 m

↳ back filling (waste rock + Mill tailings)

↳ Ht + widening upto the length of the block (other side upto S/R) along the strike

↳ Ht. of stope Raise

↳ Stope survey and marking of stowing line

↳ Back filling (waste Rock + Mill tailings)

↳ Roof Ht. of Access X cut and making ramp to access the stope ~~Unit operation~~

↳ Stoping (slicing) of 3 m ht with 9 m width ~~along the strike~~ of Sub block - 1 along the strike up to S/R

↳ Back fill of sub-block - 1

↳ Stope Survey and marking of stowing line

Date:

No.

~~Avinash (RPS) 24/4/2023~~

~~Integration of 3D-Numerical Modelling and InSAR Techniques for developing a ~~prediction model~~ land surface deformation prediction Model.~~

~~Mining induced land surface deformation prediction Model~~

~~↳ Design of literature Review~~

↳ Slicing of sub-block-2 upto the ht of 3m and width 9m along strike upto S/R

↳ ~~Be~~ Back filling of sub-block-2

↳ Stope survey and marking of stowing line

↳ Repetation of the process upto the crown pillar

↳ Access of stope from ramp-up ~~there~~ for next round of slicing (After every three round of slicing Access would be from next Access Xcut from the ramp up

No.

Unit operation in (per round of Blasting).

↳ slice Drilling → Charging & Blasting → fume clearance → Water spraying laze → Loading & Hauling → ~~Drilling~~ support drilling (back and walls) + Roof bolting (back + walls)